Homework 28.10.26 – Exercises 29–30

Gor Melkumyan

Exercise 29

Let $f = \frac{1}{n}\sum_{i=1}^{n} f_i$, and assume that each f_i is convex and L_i-smooth. Define a gradient estimator by setting $g(x) = \nabla f_i(x)$, where the index i is chosen uniformly at random. Establish a bound of the form

$$G(x,y) \le 2A\, D_f(x,y) + C,$$

and specify what A and C are. Use this to derive a convergence result for SGD using this gradient estimator.

Proof

By the definition of $G(x,y)$ and the fact that $i \sim \mathrm{Uniform}(\{1,2,\dots,n\})$,

$$G(x,y) = \mathbb{E}_i\big[\|g(x) - \nabla f(y)\|^2\big] = \frac{1}{n}\sum_{i=1}^{n} \|\nabla f_i(x) - \nabla f(y)\|^2 .$$

Using $\|a+b\|^2 \le 2\|a\|^2 + 2\|b\|^2$ (Exercise 24) we get

$$\begin{aligned} G(x,y) &= \frac{1}{n}\sum_{i=1}^{n} \big\|\big(\nabla f_i(x) - \nabla f_i(y)\big) + \big(\nabla f_i(y) - \nabla f(y)\big)\big\|^2 \\ &\le \frac{2}{n}\left(\sum_{i=1}^{n} \|\nabla f_i(x) - \nabla f_i(y)\|^2 + \sum_{i=1}^{n} \|\nabla f_i(y) - \nabla f(y)\|^2\right). \end{aligned}$$

Since each f_i is convex and L_i-smooth, we have $\|\nabla f_i(x) - \nabla f_i(y)\|^2 \le 2L_i\, D_{f_i}(x,y)$. Hence, with $L_{\max} := \max_i L_i$,

$$\begin{aligned} G(x,y) &\le \frac{4}{n}\sum_{i=1}^{n} L_i\, D_{f_i}(x,y) + \frac{2}{n}\sum_{i=1}^{n} \|\nabla f_i(y) - \nabla f(y)\|^2 \\ &\le 4L_{\max}\cdot\frac{1}{n}\sum_{i=1}^{n} D_{f_i}(x,y) + \frac{2}{n}\sum_{i=1}^{n} \|\nabla f_i(y) - \nabla f(y)\|^2 \\ &= 2\,(2L_{\max})\, D_f(x,y) + C(y), \end{aligned}$$

where we used $D_{f_i}(x,y) \ge 0$ (convexity of f_i), the linearity $D_f = \frac{1}{n}\sum_{i=1}^{n} D_{f_i}$, and

$$C(y) := \frac{2}{n}\sum_{i=1}^{n} \|\nabla f_i(y) - \nabla f(y)\|^2 = 2\,\mathrm{Var}\big[g(y)\big].$$

As a result, we get

$$A = 2L_{\max}, \qquad C = C(y) = \frac{2}{n}\sum_{i=1}^{n} \|\nabla f_i(y) - \nabla f(y)\|^2 .$$

Convergence of SGD

The estimator is unbiased:

$$\mathbb{E}_i\big[g(x)\big] = \frac{1}{n}\sum_{i=1}^{n} \nabla f_i(x) = \nabla f(x).$$

Taking $x = x^k$ and $y = x^\star$ in the bound above, and using $\nabla f(x^\star) = 0$, we get

$$G^k = \mathbb{E}\big[\left\|g^k - \nabla f(x^\star)\right\|^2 \,\big|\, x^k\big] \le 2(2L_{\max})\, D_f(x^k, x^\star) + \frac{2}{n}\sum_{i=1}^{n} \left\|\nabla f_i(x^\star)\right\|^2,$$

i.e. the AC assumption holds with

$$A = 2L_{\max}, \qquad C = 2\sigma_\star^2, \qquad \sigma_\star^2 := \frac{1}{n}\sum_{i=1}^{n} \left\|\nabla f_i(x^\star)\right\|^2.$$

If in addition f is μ-convex, Theorem 33 gives that for every stepsize $0 < \gamma \le \frac{1}{2L_{\max}}$ the iterates of SGD satisfy

$$\mathbb{E}\big[\left\|x^k - x^\star\right\|^2\big] \le (1-\gamma\mu)^k \left\|x^0 - x^\star\right\|^2 + \frac{2\gamma\sigma_\star^2}{\mu}.$$

So SGD converges linearly up to a neighbourhood of radius $\frac{2\gamma\sigma_\star^2}{\mu}$, which shrinks as $\gamma \to 0$. In particular, if $\nabla f_i(x^\star) = 0$ for all i (interpolation), then $C = 0$ and SGD converges linearly to $x^\star$.